

```

1 #include <stdio.h>
2
3 int main() {
4     float prod[12];
5     float total = 0;
6     int i, processed = 0;
7     int incentive = 0, inspection = 0;
8     int target = 5000;
9
10    printf("Enter production for 12 months (Jan-Dec).");
11    printf("\nEnter a negative value to shutdown (break). If you enter
12
13    for (i = 0; i < 12; i++) {
14        printf("Month %d: ", i + 1);
15        scanf("%f", &prod[i]);
16        if (prod[i] < 0) {
17            break;
18        }
19
20
21        if (prod[i] == 0) {
22            continue;
23        }
24
25        total += prod[i];
26        processed++;
27        if (prod[i] < 0.8 * target) {
28            inspection = 1;
29        }
30    }
31    if (processed == 0) {
32        printf("\nNo valid data processed.\n");
33        return 0;
34    }
35    float avg = total / processed;
36    float efficiency = (total / (target * processed)) * 100.0;
37    if (total > (target * processed)) incentive = 10;
38
39    printf("\n--- Results ---\n");
40    printf("Annual Production: %.2f\n", total);
41    printf("Average Monthly: %.2f\n", avg);
42    printf("Efficiency: %.2f%%\n", efficiency);
43
44    printf("Incentive: %s\n", (incentive == 10) ? "Received (10%)" : "Not Received");
45    printf("Inspection: %s\n", (inspection) ? "Marked for Inspection" : "Not Marked for Inspection");

```

5200
4800
5100
5300
....

Output

```

Annual Production: 61750.00
Average Monthly: 5145.83
Efficiency: 102.92%
Incentive: Received (10%)
Inspection: Not Required

```

```

1 #include <stdio.h>
2
3 int main() {
4     float maxCap = 500.0;
5     float load[15];
6     int maint[15] = {0};
7     for (int i = 0; i < 15; i++) load[i] = (i + 1) * 35;
8
9     maint[2] = 1;
10    maint[7] = 1;
11    for (int i = 0; i < 15; i++) {
12        if (maint[i]) {
13            printf("Sub %d: skipped (maintenance)\n", i + 1);
14            continue;
15        }
16
17        if (load[i] > maxCap) {
18            printf("Sub %d: GRID FAILURE (%.0f MW) -> break\n", i + 1,
19                load[i]);
20            break;
21        }
22
23        float loadPercent = (load[i] / maxCap) * 100.0;
24        float avail = maxCap - load[i];
25
26        if (loadPercent <= 60) printf("Sub %d: Normal, avail=%.0f MW\n", i + 1, avail);
27        else if (loadPercent <= 85) printf("Sub %d: High Load, avail=%.0f MW\n", i + 1, avail);
28        else printf("Sub %d: Critical, avail=%.0f MW\n", i + 1, avail);
29    }
30
31    return 0;
32 }

```

stdin input

Output

```

Sub 1: Normal, avail=465 MW
Sub 2: Normal, avail=430 MW
Sub 3: skipped (maintenance)
Sub 4: Normal, avail=360 MW
Sub 5: Normal, avail=325 MW
Sub 6: Normal, avail=290 MW

```

```

1 #include <stdio.h>
2
3 int main() {
4     int stock = 250000, i;
5
6     for (i = 1; i <= 25; i++) {
7         if (stock == 0) break;
8         if (stock < 8000) {
9             printf("Hospital %d: Emergency (%d)\n", i, stock);
10            stock = 0;
11            break;
12        }
13
14        stock -= 8000;
15        if (stock + 8000 >= 10000)
16            printf("Hospital %d: Sufficient\n", i);
17        else
18            printf("Hospital %d: Minimum Requirement\n", i);
19
20        continue;
21    }
22
23    printf("Remaining stock = %d\n", stock);
24    return 0;
25 }

```

stdin input

Output

```

Hospital 1: Sufficient
Hospital 2: Sufficient
Hospital 3: Sufficient
Hospital 4: Sufficient
Hospital 5: Sufficient
Hospital 6: Sufficient

```

```

1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int totalBaggage = 0;
8     int minor=0, normal=0, senior=0;
9     int addInsp=0, denied=0;
10
11     for (int i = 0; i < n; i++) {
12         int age, bag, prohibited, cancelled, emergency;
13         scanf("%d %d %d %d %d", &age, &bag, &prohibited, &cancelled, &emergency);
14
15         if (cancelled) continue;
16         if (emergency) break;
17
18         totalBaggage += bag;
19
20         if (prohibited) {
21             denied++;
22             continue;
23         }
24
25         if (bag > 30) addInsp++;
26
27         if (age < 18) minor++;
28         else if (age <= 59) normal++;
29         else senior++;
30     }
31
32     printf("TotalBaggage=%d\n", totalBaggage);
33     printf("Minor=%d Normal=%d Senior=%d\n", minor, normal, senior);
34     printf("AdditionalInspection=%d EntryDenied=%d\n", addInsp, denied);
35
36     return 0;
37 }

```

stdin input

Output

```

TotalBaggage=0
Minor=0 Normal=0 Senior=0
AdditionalInspection=0
EntryDenied=0

```